

Joel Oskarsson

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ETH Zurich AI Center, Zurich, Switzerland
Swedish (native) · English (fluent) · German (basic, A2)

Academic Career

2025 – **Postdoctoral Fellow, ETH Zurich AI Center**, Zurich, Switzerland

- Awarded 2-year fellowship in highly competitive call with a **4% acceptance rate**.
- Mentored by Torsten Hoefler (Computer Science), Siddhartha Mishra (Math), and Benedikt Soja (Space Geodesy). Topic: *Modeling the Earth System in Latent Space*.
- Funded by the Swiss AI Initiative, part of effort to develop **AI foundation models**.

2020–2025 **PhD in Computer Science, Linköping University**, 240 ECTS

- Thesis: *Modeling Spatio-Temporal Systems with Graph-based Machine Learning*
- Supervised by Fredrik Lindsten, Per Sidén, Tomas Landelius.
- Affiliated with WASP graduate school.

2015–2020 **Master's in Computer Science and Engineering, Linköping University**, 300 ECTS

- **Exchange year at ETH Zurich**, Zurich, Switzerland (2018–2019).

Research

Summary of publications

See separate document for full list of publications.

- 6 conference papers, 2 journal papers, 7 workshop papers, 3 works under review.
- 425 total citations, h-index 9 (from Google Scholar, November 2026).

Awarded Research and Career Grants

- Co-applicant for ICAIN-IGAIP project grant (320,000 CHF ≈ 345,000 EUR, **4/40 applications funded**, 2026).
  Project *AI-Powered Early Warning and Profiling for Displaced Populations*, with Christina Humer (PI, ETHZ) and the International Organization for Migration.
- Research visit scholarship (60,000 SEK ≈ 5,200 EUR), WASP Program.
- Co-author of compute grants from NSC, Sweden and CSCS, Switzerland.

Invited Talks

2026 KCDS Summer School on Graph Neural Networks, Karlsruhe Institute of Technology (KIT)
GNNs for Spatio-Temporal and Earth System Modeling

2025 **Google DeepMind**
Probabilistic Weather Forecasting with Hierarchical Graph Neural Networks

2024 RISE Learning Machines Seminars
Frontiers in Machine Learning for Weather Forecasting

European Centre for Medium-Range Weather Forecasts (ECMWF)
Graph-based Machine Learning for Weather

University of New South Wales (UNSW)
Graph-based Machine Learning for Spatio-Temporal Data, with Application to Traffic and Weather Forecasting

Selected further invited talks: WeatherGenerator Dissemination Workshop (2026); MeteoSwiss (2024); ETH Zurich (2024); Uppsala University (2024); Danish Meteorological Institute (2023).

Software Research Output

Creator and maintainer of the *Neural-LAM* **open source library** for AI weather forecasting research.

- Translates research findings into direct community impact through open-source tooling.
- **Over 300 GitHub forks and stars**, reflecting widespread usage.
- Used by multiple **international research groups**, basis for many peer-reviewed publications.
- Organizational participant in Google Summer of Code 2026, personally acting as project mentor.

Research Visits

2023 Five-week research visit to **University College London (UCL)**, London, U.K.
Hosted by Prof. Marc Deisenroth, Sustainability and Machine Learning Group.

Shorter visits: Norwegian Meteorological Institute, Oslo (2026); ECMWF, Bonn (2024); ETH Zurich + MeteoSwiss, Zurich (2024); Finnish Center for AI + Aalto University, Helsinki (2022, WASP International Study Visit).

Presentations at Scientific Events (Highlights)

- Presentations at top **machine learning conferences**, and accompanying workshops: NeurIPS (2024: poster; 2023: workshop poster), ICML (2024: workshop poster), AISTATS (2023: poster).
- Selected for **oral presentations** at the Machine Learning for Earth System Modelling Workshop, Bonn, Germany, in **three consecutive years** (2023, 2024, 2025).
- Oral presentations at the Climate Informatics Conference, Lausanne, Switzerland (2026) and the KVA Symposium on AI for Science, Stockholm (2025)

Community and Academic Service

Reviewing and Editorial Service

Area chair: ICLR (2027).

Reviewer: NeurIPS (2024; 2026 – *Top Reviewer*); ICLR (2025, 2026); AISTATS (2023, 2024, 2026); Climate Informatics (2026); Geoscientific Model Development (2026); Quarterly Journal of the Royal Meteorological Society (2026); Geophysical Research Letters (2025); SOLA (2025); Geomatics, Natural Hazards and Risk (2025, 2026); CCAI workshop (2024, 2025).

Organizing and Other Service

2026 Co-organizer of the *AI + Environment Summit 2026*, Zurich. Responsible for session on AI for weather and climate, and poster session.

2025–2026 Reviewer for applications to ETH Zurich AI Center Fellowship 2026.

2023–2025 Co-organizer of monthly online meetups for ML limited area weather forecasting researchers.

2022–2025 Organizer of the *STIMA ML Reading Group*, Linköping University.

Teaching, Supervision and Mentoring

- Supervisor of **10 master's theses** (8 LiU and 2 ETHZ, 1 ongoing) and 1 bachelor's thesis (ETHZ, ongoing).
- Mentored junior PhD students through collaborative research, co-authorship, and project guidance.
- Co- and guest-lecturer for courses *AI Center Projects in Machine Learning Research* (ETHZ) and *Time Series and Sequence Learning* (LiU).
- **Co-course-developer** of the *Foundations of Machine Learning* online course (LiU, 2021–2024).
- Pedagogical training: *Becoming a Teacher in Higher Education*, 6 ECTS (LiU, 2021).

Other teaching assistant work (2016–2024): Advanced Machine Learning (5×), Computational Statistics, Foundations of ML, Building AI, Introduction to Python, etc. All at LiU.

Media Coverage and Scientific Outreach

- *AI för prognosmeteorologer* (2025): Lecture for forecast meteorologists about AI weather forecasting.
- *Forskningsframsteg inom Maskininlärning för Väderprognoser* (2025): **Popular science article** in Polarfront, journal of the Swedish meteorological society.
- *De vill göra väderprognoser mer pålitliga – med hjälp av AI*, SVT (2024): **Television news feature**.
- *Förbättrade väderprognoser med AI*, Linköping University (2023): News article on University website.